Credal Concept Formation in Dependent Type Theory: FCA Recovery, Walley Conjugacy, and a Mizar Benchmark

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Abstract

We give a type-theoretic formalization of credal concept formation in Lean 4 that admits classical Formal Concept Analysis (FCA) as an exact, order-isomorphic special case. The contributions are: (i) subtype-based lower and upper formed-concept carriers, `LowerFormedConcept` and `UpperFormedConcept`, parameterized over a family of admissible evidence gates; (ii) an order-isomorphism recovery theorem `formedConceptOrderIsoCrispConcept` establishing that our exact lane *is* the classical FCA concept lattice; (iii) singleton-collapse equivalences that prove conservative extension of the crisp case; (iv) integration with Walley's natural extension via a four-scale conjugacy cluster on a projective credal spec, lifting the credal coordinates (lower, upper, midpoint, width-complement) into the imprecise-probability machinery the PLN $\langle s, c \rangle$ truth value compresses; (v) a first-stone empirical benchmark over an extracted slice of the Mizar Mathematical Library article `conlat_1.miz`, including a real credal witness on the `ObjectDerivation` base concept. All theorems are quoted verbatim from a public Lean 4 source with no `sorry`, `admit`, or new `axiom`. We are explicit about what is theorem-witness and what is yet-to-run experiment; §9 enumerates the empirical work required to turn the benchmark into a published discovery.

1 Introduction

Classical Formal Concept Analysis (FCA), introduced by Wille [1], gives a crisp, order-theoretic account of how concepts arise from an incidence relation $M : \mathrm{Obj} \times \mathrm{Attr} \to \{0, 1\}$. The standard objects are *formal concepts*: pairs (A, B) with $A = B^{\downarrow}$ and $B = A^{\uparrow}$ under the derivation operators. The standard structure is a complete lattice with intent/extent duality. After forty years and several thousand papers, this is the canonical mathematical framework for the question “what concepts arise from this context?” For practical concept formation over noisy or evidence-graded data, however, the binary commitment $M : \mathrm{Obj} \times \mathrm{Attr} \to \{0,1\}$ is too rigid.

Several lines of work generalize FCA to handle uncertainty. Belohlavek's fuzzy FCA [5] replaces the crisp incidence with degrees in $[0,1]$. Djouadi and Prade introduce interval-valued and possibilistic contexts [6]. Triadic FCA [4] adds a context dimension. Each gives one principled answer to “what if the relation M is uncertain?”.

This paper takes a different route. We treat the uncertainty as living *not in the relation M itself*, but in which *admissible evidence gate* converts evidence-valued cells into crisp incidence assertions. A gate is a thresholding discipline; admissible gates are those the modeller is willing to consider; concept formation under *every* admissible gate gives a *lower* carrier (“robustly formed concepts”) while concept formation under *some* admissible gate gives an *upper* carrier (“permissively formed concepts”). This matches the Walley imprecise-probability framework [8]: the gate family indexes a family of precise previsions; the lower/upper concepts are the credal envelopes of an underlying gamble. The exact (singleton-gate) case recovers classical FCA.

Three concrete contributions. The paper formalizes this picture in Lean 4 and ties it to existing PLN infrastructure as follows.

1. **Type-theoretic credal carriers (§3).** We define `LowerFormedConcept` and `UpperFormedConcept` as proper subtypes of `DualConcept`, not as wrappers around `FormedConcept`. The subtype design avoids a class of degenerate-instance traps; the upper-side witness extraction uses `Classical.choose` honestly.

2. **Order-isomorphic FCA recovery (§4).** We prove `formedConceptEquivCrispConcept` (the carrier equivalence) and `formedConceptOrderIsoCrispConcept` (the order isomorphism) against mathlib's `CrispConcept`. The recovery is not a structural resemblance: the exact lane *is* the classical FCA concept lattice as an order-theoretic object.

3. **Walley natural-extension bridge (§6).** Four parameterized conjugacy theorems at the cylinder, finite-cylinder, global, and finite-global scales of a projective local credal spec connect upper-envelope and natural-extension semantics. The credal coordinates `credalEnvelopeWidth`, `credalEnvelopeMidpoint`, `credalEnvelopeWidthComplement` are the natural $\langle s, c\rangle$-compression coordinates the PLN truth value rides on.

Empirical anchor. §7 reports a first-stone benchmark: a Python extractor walks a `.miz` article and emits an attribute-counted observation context; §7 exhibits a real credal witness on the `ObjectDerivation` base concept of `conlat_1.miz` where the concept lives in the upper carrier but not the lower, yielding $\mathrm{width} = 1$, $\mathrm{midpoint} = \frac{1}{2}$. We are explicit (§7.5) that this is theorem-witness over hand-chosen gates and one article. §8 discusses what real experiments would look like, including the most interesting one: using the upper/lower gap on a multi-article Mizar corpus to detect *formally-missing concepts* via Galois-dual reasoning.

2 Background

2.1 Classical FCA

A formal context is a triple $(\mathrm{Obj}, \mathrm{Attr}, I)$ with $I \subseteq \mathrm{Obj} \times \mathrm{Attr}$. The derivation operators $\cdot^{\uparrow} : 2^{\mathrm{Obj}} \to 2^{\mathrm{Attr}}$ and $\cdot^{\downarrow}$ form a Galois connection. A formal concept is a pair (A, B) with $A^{\uparrow} = B$ and $B^{\downarrow} = A$. The set of formal concepts ordered by $(A, B) \leq (A', B') \iff A \subseteq A'$ is the *concept lattice* of the context. The main theorem of FCA [1, 2] is that every finite lattice arises this way.

Algorithmic FCA tooling (NextClosure, ConExp, FCAbedrock, Lattice Miner) computes concept lattices in time proportional to lattice size, not $2^{|\mathrm{Obj}|}$. For practical contexts the lattice stays small. The crisp commitment, however, is brittle: a single attribute-cell flip can move concepts in and out of the lattice discontinuously.

2.2 Imprecise probability and Walley's natural extension

Walley [8] replaces a single probability measure with a *credal set* of admissible precise previsions. A lower prevision $\underline{P}$ is *coherent* when it avoids sure loss and satisfies superadditivity and positive homogeneity. The *natural extension* of a coherent lower prevision is the greatest coherent lower prevision dominated by it; its conjugate $\overline{P}(X) = -\underline{P}(-X)$ is the corresponding upper prevision. The *lower envelope* of a credal set $\mathcal{C}$ on a gamble X is $\inf_{P \in \mathcal{C}} P(X)$; the *upper envelope* is sup. Walley's representation theorem links the two notions: every coherent lower prevision is the lower envelope of its set of dominating precise previsions, and conversely.

Belohlavek's fuzzy FCA [5] and Djouadi–Prade's interval-valued FCA [6] are predecessors for treating FCA-style closures under uncertainty. They do not bridge into Walley's natural-extension calculus, and do not formalize a conservative reduction to crisp FCA in dependent type theory.

2.3 PLN truth values

Probabilistic Logic Networks [10] represent the strength and confidence of an inheritance relation as a two-number tuple $\langle s, c\rangle$, where $s \in [0, 1]$ is interpreted as a strength-like point estimate and $c \in [0, 1]$ as a confidence-like inverse-imprecision. A companion characterization paper [11] identifies $\langle s, c\rangle$ as a two-number compression of an underlying credal set, with three layered slots where the compression admits genuine degrees of freedom. The present paper provides the *concept-formation* side of that picture: how credal carriers, FCA recovery, and Walley conjugacy together justify a midpoint/width-complement reading of $\langle s, c\rangle$.

3 The credal carriers

The exact (precise) concept-formation surface in our framework is given by the `ObservationSurface` machinery and its `finiteConceptFamily` operator; see [13] for the public-API discussion. Codex's `ConceptOntology/Formation.lean` provides the operator and a correctness theorem $A \in \mathrm{finiteConceptFamily}(S, G, \sigma)$ iff A is closed under $\mathrm{crispRelation}(G, S.\mathrm{aggregate}(\sigma))$. The credal extension lives one level up.

Carriers as subtypes. The credal carriers are not wrappers; they are subtypes whose proof field carries genuine content.

Definition 1 (`LowerFormedConcept` and `UpperFormedConcept`, `ConceptOntology/CredalFormation.lean:143,148`).

```
abbrev LowerFormedConceptΓ
    ( : Gate → EvidenceGate Q) (M : Obj → Attr → Q) :=
  { A : DualConcept Obj Attr // A   lowerConceptFamily Γ M }

abbrev UpperFormedConceptΓ
    ( : Gate → EvidenceGate Q) (M : Obj → Attr → Q) :=
  { A : DualConcept Obj Attr // A   upperConceptFamily Γ M }
```

The membership predicate `lowerConceptFamily` is "A is a formed concept under *every* gate $\Gamma(g)$" and `upperConceptFamily` is "A is a formed concept under *some* gate." The carriers' proof field $A.2$ certifies $A \in \mathrm{lowerConceptFamily}(\Gamma, M)$: an honest dependent proof obligation, not a definitional unfolding.

Why subtype, not wrapper. The carrier design avoids a degenerate pattern that would render the credal layer vacuous: defining a "credal concept" as a *renamed* `FormedConcept` and proving the recovery theorem by `rfl`. The present subtype design forces the recovery theorem to do real work because the carrier types are genuinely distinct, even though their meaning sets are related.

Working with the proof field. The single-gate projection `lowerFormedConceptAt` uses the universal-gate proof concretely:

```
def lowerFormedConceptAtΓ
    ( : Gate → EvidenceGate Q) (M : Obj → Attr → Q)
    (g : Gate) :
    LowerFormedConcept Γ M → AbstractInheritance.FormedConcept Γ( g)
        M :=
  fun A => A.1, A.2 g
```

(`ConceptOntology/CredalFormation.lean:198`.) The upper-side witness extraction uses `Classical.choose`:

```
noncomputable def upperFormedConceptWitnessΓ
    ( : Gate → EvidenceGate Q) (M : Obj → Attr → Q) :
    UpperFormedConcept Γ M →Σ
       g : Gate, AbstractInheritance.FormedConcept Γ( g) M := by
  classical
  intro A
  refine  Classical.choose A.2, ? _
  exact  A.1, Classical.choose_spec A .2
```

(line 205). Both projections preserve the underlying dual-concept value unchanged; this is recorded as the `lowerFormedConceptAt_val` and `upperFormedConceptWitness_val` simp lemmas.

4 FCA recovery

Codex's `ConceptOntology/FCARecovery.lean` module makes the FCA recovery explicit. The 74-line file proves two theorems and one corollary; we quote the first verbatim and sketch the second.

Theorem 2 (Carrier equivalence, `ConceptOntology/FCARecovery.lean:27`).

```
noncomputable def formedConceptEquivCrispConcept
    (G : EvidenceGate Q) (M : Obj → Attr → Q) :
    AbstractInheritance.FormedConcept G M  CrispConcept G M where
  toFun A :=
    DualConcept.toConcept A.1
      (AbstractInheritance.formedConcept_isClosed G M A)
  invFun c :=
    DualConcept.ofConcept c,
      (AbstractInheritance.mem_finiteConceptFamily_iff G M _).2
        (DualConcept.isClosed_ofConcept c)
  left_inv A := by
    apply Subtype.ext
    exact DualConcept.ofConcept_toConcept A.1
      (AbstractInheritance.formedConcept_isClosed G M A)
  right_inv c := by
    exact DualConcept.toConcept_ofConcept c
```

The carrier equivalence promotes a formed-concept proof to a classical formal-concept pair (A, B) via the `DualConcept.toConcept` construction, and inverts it by `DualConcept.ofConcept`. Both inverse laws are honest sigma-extensionality lemmas, not `rfl`.

Theorem 3 (Order isomorphism, `ConceptOntology/FCARecovery.lean:44`).

```
noncomputable def formedConceptOrderIsoCrispConcept
    (G : EvidenceGate Q) (M : Obj → Attr → Q) :
    AbstractInheritance.FormedConcept G M o CrispConcept G M where
  toEquiv := formedConceptEquivCrispConcept G M
  map_rel_iff' := by
    intro A B
    simpa using
```

```
    (DualConcept.toConcept_le_iff A.1 B.1
      (AbstractInheritance.formedConcept_isClosed G M A)
      (AbstractInheritance.formedConcept_isClosed G M B))
```

This is the headline claim of the recovery layer: the exact formed-concept order is the classical FCA concept-lattice order, as an order isomorphism. It is not merely the case that the carriers are equipotent; the order structure transfers exactly. A corollary at line 64, `formedConcept_inherits_iff_crispConcept_le`, packages the implication for downstream consumers who want the inheritance-style direction.

4.1 Singleton-collapse conservative extension

The credal layer collapses to the exact layer in the singleton-gate case. We formalize this via two type equivalences:

Theorem 4 (Lower singleton equivalence, `ConceptOntology/CredalFormation.lean:275`).

```
noncomputable def lowerFormedConceptSingletonEquiv
    (G : EvidenceGate Q) (M : Obj → Attr → Q) :
    LowerFormedConcept (Gate := PUnit) (fun _ => G) M
      AbstractInheritance.FormedConcept G M
```

The companion `upperFormedConceptSingletonEquiv` at line 292 is the upper-side twin. Both proofs use the singleton iff lemmas `mem_lowerConceptFamily_singleton_iff` and `mem_upperConceptFamily_singleton_iff`, with `casesA; rfl` discharging the inverse laws. The constructions are honest equivalences between type-distinct objects (the credal subtype *with* `Gate:=PUnit` versus the bare `FormedConcept`).

What conservative extension means here. When the gate family has a single element, every `LowerFormedConcept` is the same as a `FormedConcept` under that gate, and the same for `UpperFormedConcept`. Composed with Theorem 3, this gives the chain

$$\text{Credal layer at singleton gate} \xrightarrow{\text{Thm. 4}} \text{Exact } \texttt{FormedConcept} \xrightarrow{\text{Thm. 3}} \text{Classical FCA concept lattice.}$$

Both arrows are order isomorphisms. The credal layer is a conservative refinement, not a sideways alternative.

5 Inheritance and transport

The credal carrier composes cleanly with the existing inheritance, closure, and ontology-growth infrastructure landed in earlier passes (`Logic/EmpiricalIntensionalFactorGraphBridge.lean`, `Logic/CredalConceptFixpointClosureBridge.lean`, `Logic/CredalConceptOntologyGrowthBridge.lean`).

5.1 Lower-credal inheritance table

The credal inheritance table mirrors the precise version at line 1644 but operates on `LowerFormedConcept`:

Definition 5 (`lowerCredalConceptInheritanceTable`, `EmpiricalIntensionalFactorGraphBridge.lean:1778`).

```
noncomputable def lowerCredalConceptInheritanceTableΓ
    ( : Gate → EvidenceGate Q)
    (M : Obj → Attr → Q)
    (subConcept superConcept : LowerFormedConcept Γ M) :
    FiniteWitnessFeatureTable :=
  Interpretation.toFiniteWitnessFeatureTable
    (lowerFormedConceptInterpretation Γ M)
    subConcept superConcept
```

Three exactness theorems follow: the prior at line 1787, the strength at line 1804, and the log-ratio at line 1822. Each proves the credal interpretation's read of the corresponding quantity equals the table read, via `simpa` from the underlying precise lemma. The packaged conjunction `lowerFormedConceptInheritance_exact_via_table` at line 1985 is the credal analogue of the line-1644 precise theorem.

5.2 Closure transport

The credal inheritance survives closure under WM-style rule sets:

Theorem 6 (Threshold validity through closure, `CredalConceptFixpointClosureBridge.lean:71`). *Let seed be a set of `LowerFormedConceptPair`, R a `RuleSet`. If a query-encoding encode exposes the supported lower-formed extensional-inheritance slice at strength level (the `hEncode` hypothesis), and every seed obligation is above threshold τ according to the exact generated table, then the encoded seed query set is threshold-valid in the world model. The companion `leastRuleClosure_thresholdValid_of_exactTableStrength` at line 114 shows the same obligations stay threshold-valid after applying any state-indexed WM consequence rule set.*

5.3 Ontology-growth transport

The credal inheritance is stable under ontology growth, provided the agreement region is interaction-closed and the two MLN specs satisfy the `PaperUniformSmallTotalInfluence` (Dobrushin-style) bound:

Theorem 7 (Stable under ontology growth, `CredalConceptOntologyGrowthBridge.lean:87`). *If a lower-formed-concept seed family is threshold-valid in MLN world-model state μ_1 and every encoded seed query is supported inside an agreement region Γ, then the same seed family remains threshold-valid after ontology growth to the agreeing spec on μ_2. The companion line 145 theorem extends the result through least-rule closure, and lines 212/280 push the result into perspectival availability/admissibility.*

Theorems 6 and 7 together demonstrate that the credal carrier is not a parallel stack. It integrates with the existing world-model semantics: credal inheritance obligations transport through both *logical* closure (rule-set application) and *ontological* closure (Dobrushin-bound growth). This is a conditional transport theorem: it does not assert that arbitrary ontology growth is stable, but that stability follows under the stated agreement-region and Dobrushin-style hypotheses.

6 Credal truth coordinates and the Walley bridge

The credal layer reads $\langle s, c\rangle$-like coordinates off the lower/upper envelopes through definitions in `ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean`.

6.1 The three coordinates

Definition 8 (Credal envelope coordinates, `ProjectiveCredal.lean`).

```
noncomputable def credalEnvelopeWidth
    (C : CredalPrevisionSet Ω) (X : Gamble Ω) :   :=
  upperEnvelope C X - lowerEnvelope C X

noncomputable def credalEnvelopeWidthComplement
    (C : CredalPrevisionSet Ω) (X : Gamble Ω) :   :=
  1 - credalEnvelopeWidth C X

noncomputable def credalEnvelopeMidpoint
    (C : CredalPrevisionSet Ω) (X : Gamble Ω) :   :=
  (lowerEnvelope C X + upperEnvelope C X) / 2
```

The docstrings are explicit about the intended PLN reading: `credalEnvelopeWidth` is the imprecision coordinate that "PLN confidence-like displays compress"; `credalEnvelopeMidpoint` is the point-estimate coordinate that "PLN strength-like displays compress."

The display map used in this paper is the canonical midpoint / width-complement projection for this credal reading of PLN truth values:

$$\langle s, c\rangle \longleftarrow \big(\texttt{credalEnvelopeMidpoint}\, C\, X,\ \texttt{credalEnvelopeWidthComplement}\, C\, X\big).$$

Singleton-recovery: when `credalSetDetermines` $C\,X$ holds, the lower and upper envelopes coincide and the midpoint reads the precise value while the width-complement reads 1, recovering the precise PLN truth value. This is the "three slots of freedom" picture of [11] viewed from the concept-formation side.

The same display lane is now packaged for credal inheritance pairs in `EmpiricalIntensionalFactorGraphBridge.lean`. The gamble `credalInheritanceGamble` uses the independent gate index `Gate×Gate`, so it matches the existing predicate `credalInheritanceJudgment`: the subconcept and superconcept each need upper support, but not necessarily under the same admissible gate. The global readout theorems `globalEnvelopeWidth_credalInheritanceGamble_eq`, `globalEnvelopeWidthComplement_credalInheritanceGamble_eq`, and `globalEnvelopeMidpoint_credalInheritanceGamble_eq` prove the intended PLN coordinates: `credallyImpreciseInheritance` gives width 1 and width-complement 0, while robust precision gives width-complement 1. The review-facing bundle is `CredalInheritanceTruthCoordinateCrown`.

6.2 The Walley conjugacy cluster

A four-theorem cluster lands the conjugacy $\overline{P}^{*} = -\underline{P}_{\text{nat. ext.}}(-\cdot)$ at four scales of the projective credal spec: cylinder, finite-cylinder, global, finite-global. We quote the global one verbatim; the cylinder one follows the same pattern at a smaller scale.

Theorem 9 (Global upper-envelope/natural-extension conjugacy, `ProjectiveCredal.lean:6871`).

```
theorem globalUpperEnvelopePrevision_conjugate_eq_naturalExtension
    (S : ProjectiveLocalCredalSpec Window Global)
    (hNonempty : S.hasCompatibleCompletion)
    (hBdd :   X : Gamble Global,
      BddAbove ((fun P : PrecisePrevision Global => P X) ''
        S.projectiveLimitCredalSet))
    (X : Gamble Global) :
    (S.globalUpperEnvelopePrevision hNonempty hBdd).conjugate X =
      S.globalNaturalExtension X
```

The proof is four lines: invoke `upperEnvelopePrevision_conjugate_eq_lowerEnvelope` at the credal-set level on `S.projectiveLimitCredalSet`, with the spec's nonempty/compatibility hypothesis as input. The three siblings live at lines 6268 (cylinder), 6523 (finite-cylinder), and 7065 (finite-global), with the same proof shape, scaled hypotheses, and matching siblings going the other direction (`globalNaturalExtension_conjugate_eq_upperEnvelope`).

What the cluster buys. Each conjugacy theorem is a Walley-canonical compatibility lemma: the spec's packaged upper-envelope prevision (a thing of type `UpperPrevision`) and the spec's packaged natural extension (a thing of type `LowerPrevision`) are conjugate. This is the load-bearing identity tying the exact-natural-extension calculus to the credal-set envelope calculus on the projective-credal infrastructure. The credal coordinates of Definition 8, when read off `S.projectiveLimitCredalSet`, are the standard Walley coordinates of the natural extension.

7 The Mizar benchmark

We exhibit the first concrete benchmark: an extractor for Mizar Mathematical Library articles, a generated context, and a credal witness with computed coordinates.

7.1 The extractor

`scripts/concept_ontology/extract_mizar_article_context.py` (330 lines) walks a Mizar source file and emits a JSON observation context. Rows are classified into four kinds (`definition`, `registration`, `notation`, `theorem`); attributes are extracted from the article's mode/attribute declarations and predicate uses. Output schema:

```
{
  "article":    string,
  "attributes": [ string ],
  "rows": [
    { "object":   string,
      "kind":     "definition"|"theorem"|"registration"|"notation",
      "index":    int,
      "preview":  string,
      "counts":   { attribute -> int } } ]
}
```

The extractor is deliberately small and conservative: it does not interpret Mizar's full grammar, only enough to populate the attribute-count cells. The output is then loaded by a generated Lean module (`Generated/MizarConlat1.lean`) into the `BinaryFcaBenchmarkContext` surface from `BenchmarkControl.lean`.

7.2 Working slice: `conlat_1.miz`

`conlat_1.miz` (Wojciechowski, 2002, on Concept Lattices, 3260 source lines) is chosen for two reasons: it is a concept-lattice article, so the predicate vocabulary is FCA-aligned, and it is well-scoped (mid-size, no heavy preface). The extractor produces 14 rows over 9 attributes:

Attribute	**Role**
`Attribute`	FCA primitive
`AttributeDerivation`	Galois operator (attribute side)
`ContextStr`	2-sorted context structure
`FormalContext`	Article's main concept-context object
`Function`	Underlying carrier function
`Information`	Incidence relation
`ObjectDerivation`	Galois operator (object side)
`Subset`	Stratification primitive
`is-connected-with`	Incidence predicate alias

Row-kind histogram: 6 definitions, 5 theorems, 2 registrations, 1 notation.

7.3 Toy control: the flying-family example

`ConceptOntology/BenchmarkControl.lean` provides a small toy concept-context (animals with traits) used to exercise the FCA recovery layer with concrete witnesses. The positive witness `flyingFamilyConcept_mem_exact` at line 238 shows the flying-family concept is in the exact concept family; the negative witness `batOnlyFlyingConcept_not_mem_exact` at line 246 shows a non-closed pair is correctly rejected; the lower-credal exclusion `flyingFamilyConcept_not_mem_lower` at line 265 shows the lower-credal carrier correctly rejects the flying-family concept under a strict-gate family that requires two supporting tokens. This is the controlled benchmark surface against which the Mizar layer is instantiated.

7.4 The credal witness on `conlat_1`

`ConceptOntology/MizarBenchmark.lean` instantiates the `BinaryFcaBenchmarkContext` surface on the extracted `conlat_1` context. A two-element gate family `MizarGate` := {`loose`, `strict`} is defined: `loose` = the exact (positive-support) gate; `strict` = the positive-threshold-2 gate (requires two supporting tokens per cell).

The article's `ObjectDerivation` attribute concept is encoded as `objectDerivationLooseConcept` (line 67). The credal witnesses are:

- `objectDerivationLooseConcept_mem_upper` (line 127): the concept is in `upperMizarConceptFamily` (it forms under the `loose` gate).
- `objectDerivationLooseConcept_not_mem_strict` (line 131): it is not closed under the `strict` gate's incidence relation.
- `objectDerivationLooseConcept_not_mem_lower` (line 138): therefore it is not in `lowerMizarConceptFamily`.

The three computed coordinate theorems (lines 147, 160, 173) follow from the upper-only-membership pattern:

$$\text{width} = 1, \qquad \text{midpoint} = \frac{1}{2}, \qquad \text{width-complement} = 0.$$

These are the maximum-imprecision coordinates one would expect for a concept that lives in upper-but-not-lower under a binary gate family: the lower envelope reads 0, the upper reads 1, the midpoint is $\frac{1}{2}$, and the width-complement (i.e., PLN-style confidence) is at zero.

7.5 Honest accounting

The reader should not over-interpret the Mizar witness. Specifically:

1. **One article, hand-picked gates.** The witness is computed on a single `.miz` file with two artificial gates (positive-threshold-1 versus positive-threshold-2). The credal coordinates are not measured against natural gate variation in the Mizar Library; they are demonstrated against codex-chosen thresholds.

2. width = 1 **is structural, not empirical.** Under a binary gate family where one gate says yes and the other says no, the lower envelope reads 0 and the upper reads 1, giving width exactly 1. The witness verifies the calculus, not the substantive credal claim about MML.

3. **External FCA checking is a separate gate.** The order-isomorphism Theorem 3 is the theorem-side claim that our exact lane is classical FCA. An empirical cross-check against ConExp or LatticeMiner on the extracted context is straightforward (the JSON schema is convertible to standard `.cxt` format) but is not the load-bearing result of this paper.

4. **Gate-family choice is itself a research question.** The curated-family pilot in `artifacts/concept_ontology/mizar_family/credal_threshold_summary.json` now runs threshold gates over 13 lattice/FCA-adjacent MML articles. In that pilot, all 13 articles have more than one induced concept family and at least one upper-minus-lower concept under the sampled count thresholds; the aggregate unstable-concept count is 917. This is evidence that the extraction and credal scoring harness is live, not evidence that the sampled threshold gates are the right scientific gate family. On classical binary FCA contexts (`livingbeings_en.cxt`, `animals_en.cxt`), the same threshold gates degenerate: threshold ≥ 1 recovers the original context and threshold ≥ 2 yields a near-null one, leaving the credal envelope structureless. Row-drop and column-drop *observation* gates do produce meaningful lower/upper families on those binary contexts. The diagnostic is that the credal layer applies not whenever "we have a dataset" but whenever "we have a meaningful family of admissible gates for that dataset." Choosing the gate family to match the evidence type is part of the empirical work; the theory does not automate it.

What this benchmark *does* demonstrate is that the formal infrastructure runs end-to-end on real Mizar source: extractor → context → credal layer → coordinates, and that the same machinery can now be run as a small article-family pilot. It is the first stone of a real benchmark, not yet a benchmark result. §8 discusses the experiments that would turn this into one.

7.6 The semantic-feature lane: status

Roadmap item (f) below — extracting *semantic* rather than lexical features — is now operational, and reporting its current state at true weight is more useful than deferring it. For a curated slice of the `LATTICE2` article (`MPTP/chainy` export, theorems `t14`–`t25`) we extract, per theorem, a Craig interpolant on the entailment $\Gamma \vdash \varphi$ (Vampire symbol elimination) and a finite model of the satisfiable premise fragment (Vampire FMB [17]); the predicate/function symbols of each artifact become the attributes of an FCA context, and the credal gate family runs as in §7.4. Three findings, each stated at its true weight.

(1) A reusable schema synthesizer. A single recognizer-based procedure replaces the earlier per-cluster extractors: given a theorem cluster it emits a parameterized schema (a theorem-with-blanks) or refuses with "no common schema." On `t14`–`t17` it recovers the left-identity/extremum schema — *in a bounded lattice, the unique left identity of meet (resp. join) is the top (resp. bottom)* — with the operation and the extremal element as the two parameter slots, and it *refuses* the mixed `t14`–`t19` cluster rather than fabricating a common template. We stress that this schema is an elementary lattice fact: it witnesses that the engine recovers a genuine unification from raw first-order export, and is *not* offered as a mathematical result. On the strip-mined tail (`t20`–`t25`) the synthesized families drift toward library *encoding* regularities (self-relations of the `BINOP_1` operation functor) rather than lattice mathematics — a faithful signal that elementary `LATTICE2` is exhausted as a source of interesting structure.

(2) A cross-role bridge. The cleanest non-degenerate concept is the pair `t15`+`t18` and its dual `t17`+`t19`: a left-identity-uniqueness theorem and a unity-identification theorem that target the *same* extremal object through the *same* selector. This is a readable concept-family bridge between two theorem forms, confirmed by direct FCA closure, and is not a solver-timeout artifact.

(3) No gate yet perturbs the mathematics. We ran three gate families. Premise-neighborhood subsetting is a no-op (the prover already uses only the neighborhood). Whole-symbol operator ablation is destructive: the resulting instability is dominated by `failure:timeout` — proof *failure*, not changed content. A non-deletion "right-partition" family — moving a support record across the Craig interpolation cut instead of deleting it — is solver-robust (zero timeouts) and does change the interpolant vocabulary, but it perturbs the *feature* (which symbols are forced into the shared interpolation vocabulary), *not the theory*: every gate leaves the theorem's truth untouched. It is a feature-stability diagnostic, not a conjecture generator. The gate family that would generate conjectures is stated as item (f).

8 What real experiments would look like

We are explicit about which experiments would move the benchmark from infrastructure-witness to scientific result.

(a) External FCA cross-check. Convert `Generated/MizarConlat1.json` to a standard FCA `.cxt` file, compute the concept lattice in ConExp 1.1.4 or FCAbedrock, and compare against the Lean-computed exact concept family for size, edges, and selected concept witnesses. The order-isomorphism theorem predicts perfect agreement. ∼2 days of engineering. This is sanity, not novelty.

(b) Multi-article scale. A 13-article lattice/FCA-family pilot is now materialized in `artifacts/concept_ontology/mizar_family/`: the extractor writes per-article JSON and `.cxt` files, the threshold-gate summary records non-trivial upper/lower splits in every article of the pilot, and the aggregate unstable-concept count is 917. The next scale target is 50–100 MML articles. The mathematical library has ∼1500 articles total; the `chainy` filter of MPTP [14] already produces well-scoped subsets. The extractor's per-article extraction is fast; loading all articles into one Lean module is not, so the harness needs an article-batched scoring layer rather than a single monolithic context. The output is a real Concept × Article presence matrix.

(c) Article-derived gates. Replace the hand-picked threshold gates of §7.4 with article-derived gates (each MML article becomes one admissible gate; the credal carriers then witness "concepts robust across the article portfolio" versus "concepts attested in at least one article"). This is the move from synthetic to semantic credal variation.

(d) Galois-dual ghost hunt. FCA theory predicts every closure operator has a Galois dual; every concept-lattice node has a corresponding co-lattice node. `conlat_1.miz` itself exhibits this: `ObjectDerivation` and `AttributeDerivation` are Galois-dual operators. If the multi-article scan identifies concepts in Upper \ Lower where the FCA-dual concept is expected but not formally registered in some articles, those candidates are predictions of "formally missing concepts" in MML that the credal layer surfaces via duality reasoning. The first narrow lexical scan, `artifacts/concept_ontology/mizar_family/duality_ghosts.json`, reports three one-sided `Top/Bottom` duality-mention candidates (`conlat_1`, `yellow_2`, `waybel_0`); this is a triage list, not a semantic confirmation of missing concepts. Verifying a small number of such predictions against domain experts (Mizar maintainers, recent MML changelog) would constitute a published discovery in the FCA-on-formal-math territory. We give this a ∼65% probability of being feasible at the formalization layer; ∼50% of producing at least one publishable discovery.

(f) Conjectures live in the prove–refute gap. The semantic lane of §7.6 is operational but has not yet surfaced a non-trivial conjecture, because every gate built so far either leaves the theory fixed (right-partition) or destroys provability without establishing falsity (operator ablation). The gate that *generates* conjectures is hypothesis weakening with a dual oracle: drop a hypothesis from a synthesized schema, then ask both a prover (does the weakened schema still prove?) and a model builder (does its negation admit a finite countermodel?). Three outcomes partition the result: *proved* (the hypothesis was unnecessary — a generalization theorem); *refuted* (the hypothesis was essential — a sharpness witness); and *neither, in budget* — the genuine open conjecture. The credal layer ranks candidates by gate-stability, and a companion Markov-logic world-model formalization (which proves queryStrength = queryProb, i.e. the PLN strength a world model assigns to a query equals its Markov-network probability) attaches a truth-likelihood to the third class. The honest discriminating questions then are whether the credal coordinates beat a syntactic baseline (ENIGMA-style symbol features, raw overlap) and whether the truth-likelihood is *calibrated* against hammer-provability on decidable held-out conjectures. Crucially, this experiment requires a substrate richer than elementary lattice theory — where the prove–refute gap is empty within budget — such as the `WAYBEL` continuous-lattice articles, or cross-article schema transfer in which a schema proven in one domain becomes a conjecture in another.

(g) Annotator-disagreement complement. Walley-canonical credal benchmarks come from real annotator disagreement: CIFAR-10H [18] (51 labels per image), ChaosNLI [19] (100 annotators per item). With appropriate object-attribute-gate reframing (classes as attributes, annotators as gates, ~ 200–500 stratified items), these become tractable Lean benchmarks. This is the headline credal-empirical complement to the formal-mathematical Mizar benchmark of (b)–(d). Out of scope for the present paper.

9 Related work

Classical FCA. Wille's [1] original triadic-context-free treatment; the canonical reference Ganter–Wille [2]; the survey volume [3]. Mathlib's `Mathlib/Order/Concept.lean` provides a Lean 4 formalization of the crisp concept-lattice carrier. The present work's recovery theorem (Thm. 3) bridges our exact concept layer into that mathlib artifact; we do not duplicate it.

Generalized FCA. Belohlavek's fuzzy FCA [5] replaces the crisp incidence with degrees in $[0, 1]$. Djouadi and Prade [6] introduce interval-valued and possibilistic contexts. Murai–Mizumoto–Tahani [7] treat the topic from a possibility-theoretic angle. Triadic FCA [4] adds a context dimension. None of these works formalize a conservative reduction to crisp FCA in dependent type theory, nor integrate with Walley's natural-extension calculus through a projective credal spec, nor connect to a PLN-style truth-value calculus.

Walley framework. The standard reference is [8]; the modern handbook is [9]. Natural extension's role as the canonical coherent extension and as the conjugate of the upper envelope is treated there.

Lean formalizations of imprecise probability. The `Mettapedia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean` module is a multi-thousand-line Lean 4 formalization of credal sets, lower/upper previsions, natural extension, and projective credal specs, on which the present paper relies. To our knowledge it is the largest extant Lean 4 formalization in this area.

PLN. Goertzel–Iklé–Ari–Heljakka [10] introduce the two-number truth value; the companion characterization paper [11] formalizes its three slots of degrees of freedom over the same projective credal spec used here. Sibling formalization papers `xiPLN.tex` and `nuPLN.tex` establish the broader PLN-on-Mettapedia program.

Mizar. Bancerek et al. [15] give the canonical reference for the Mizar Mathematical Library; Urban [14] introduces MPTP. Premise-selection work over MML/MPTP includes Jakubův–Urban ENIGMA [16] and the present project's sibling PLN-kNN-NB [12]. The Mizar extractor presented here is, to our knowledge, the first FCA/credal-layer formalization of MML article structure; prior MML-engagement work has been ATP-centric.

Automated theory formation and theory exploration. The concept-formation → conjecture → prove → counterexample cycle is not new. Colton, Bundy, and Walsh's HR [20] forms concepts and conjectures over data tables and dispatches them to a prover and a model generator, scored by deterministic interestingness measures; Colton and Wagner [21] applied FCA to mathematical discovery over the OEIS and Graffiti conjectures. Theory-exploration systems QuickSpec [22] and Hipster [23] synthesize and filter equational conjectures for proof assistants. Our *cycle shape* and our *use of FCA for mathematical discovery* are therefore not the novel elements, and we do not claim them as such. The delta is narrow and specific: a Walley-style credal lower/upper family *indexed by a gate family* over *prover-derived* semantic artifacts (Craig interpolants, finite models), with a Lean-4 exact/credal foundation and a PLN truth-coordinate reading. We likewise claim no novelty for uncertainty-aware FCA in general (cf. the generalized-FCA paragraph) nor for automated conjecturing in general.

10 Conclusion and open work

Recap of contributions. (i) Type-theoretic credal carriers for concept formation as proper subtypes; (ii) FCA recovery as a Lean 4 order-isomorphism theorem against mathlib's concept-lattice infrastructure; (iii) singleton-collapse conservative-extension theorems for both lower and upper carriers; (iv) integration with Walley's natural extension via a four-scale conjugacy cluster on a projective credal spec under the compatible-completion and boundedness hypotheses of §6.2; (v) a first-stone Mizar benchmark with a real credal witness on `conlat_1.miz`'s `ObjectDerivation` concept.

Open work. The upper-side inheritance table and the basic two-sided credal inheritance predicates have now landed: `upperCredalConceptInheritanceTable`, `credalInheritanceJudgment`, `credallyPreciseInheritance`, and `credallyImpreciseInheritance`. These predicates now have a PLN truth-coordinate package through `CredalInheritanceTruthCoordinateCrown`, including midpoint and width-complement readouts for inheritance. A 13-article Mizar-family pilot and first narrow duality-ghost scan are now materialized as artifacts. The remaining work is to scale the pilot to a larger, less curated MML slice, replace convenience threshold gates with article-derived semantic gates, and validate any ghost candidates against Mizar-domain expertise.

What the paper is not. The paper is not a benchmark result. It is the formal-infrastructure writeup of a layer that can support a benchmark result. The load-bearing scientific question — "can the credal layer surface formally-missing concepts in real mathematical libraries?" — requires the multi-article scale of §8(b)–(d) and is left open.

Hygiene. All theorems quoted in this paper are verbatim from the public Lean 4 source as of the date listed on the title page. A grep across the cited files (`ConceptOntology/CredalFormation.lean`, `ConceptOntology/FCARecovery.lean`, `ConceptOntology/BenchmarkControl.lean`, `ConceptOntology/MizarBenchmark.lean`, `EmpiricalIntensionalFactorGraphBridge.lean`, `CredalConceptFixpointClosureBridge.lean`, `CredalConceptOntologyGrowthBridge.lean`, `ProjectiveCredal.lean`) for `sorry`, `admit`, `axiom`, and `bytrivial` returns zero hits. Axiom checks on the load-bearing theorems return the mathlib baseline {`propext`, `Classical.choice`, `Quot.sound`}.

References

[1] R. Wille. Restructuring lattice theory: an approach based on hierarchies of concepts. In *Ordered Sets* (I. Rival, ed.), Reidel, 1982, pp. 445–470.

[2] B. Ganter and R. Wille. *Formal Concept Analysis: Mathematical Foundations*. Springer, 1999.

[3] B. Ganter, G. Stumme, and R. Wille (eds.). *Formal Concept Analysis: Foundations and Applications*. LNAI 3626, Springer, 2005.

[4] R. Wille. The basic theorem of triadic concept analysis. *Order*, 12(2):149–158, 1995.

[5] R. Belohlavek. Fuzzy Galois connections. *Mathematical Logic Quarterly*, 45(4):497–504, 1999.

[6] Y. Djouadi and H. Prade. Interval-valued fuzzy formal concept analysis. In *ISMIS 2009*, LNCS 5722, Springer, 2009, pp. 592–601.

[7] T. Murai, M. Mizumoto, and N. Tahani. Possibility theory and concept lattices. In *Proc. IFSA 1989*, 1989.

[8] P. Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman & Hall, 1991.

[9] T. Augustin, F. P. A. Coolen, G. de Cooman, and M. C. M. Troffaes. *Introduction to Imprecise Probabilities*. Wiley, 2014.

[10] B. Goertzel, M. Iklé, I. F. Goertzel, and A. Heljakka. *Probabilistic Logic Networks: A Comprehensive Framework for Uncertain Inference*. Springer, 2008.

[11] Mettapedia Working Notes. Credal degrees of freedom for PLN truth values. Companion paper, this collection, 2026.

[12] Mettapedia Working Notes. PLN-kNN-NB on extended MPTP 5k. Companion paper, this collection, 2026.

[13] GPT-Pro 5.4. WM-PLN concept formation grounding guidance. `GPT-Pro_Responses/WM-PLN/wmpln_concept_formation_grounding_guidance.md`, 2026.

[14] J. Urban. MPTP – Motivation, implementation, first experiments. *Journal of Automated Reasoning*, 33(3-4):319–339, 2004.

[15] G. Bancerek, C. Bylinski, A. Grabowski, A. Kornilowicz, R. Matuszewski, A. Naumowicz, K. Pak, and J. Urban. The role of the Mizar Mathematical Library for interactive proof development in Mizar. *Journal of Automated Reasoning*, 61(1-4):9–32, 2018.

[16] J. Jakubův and J. Urban. Hammering Mizar by learning clause guidance. In *ITP 2019*, LIPIcs 141, 2019, pp. 34:1–34:8.

[17] G. Reger, M. Suda, and A. Voronkov. Finding finite models in multi-sorted first-order logic. In *SAT 2016*, LNCS 9710, Springer, 2016, pp. 323–341.

[18] J. C. Peterson, R. M. Battleday, T. L. Griffiths, and O. Russakovsky. Human uncertainty makes classification more robust. In *ICCV 2019*, pp. 9617–9626.

[19] Y. Nie, X. Zhou, and M. Bansal. What can we learn from collective human opinions on natural language inference data? In *EMNLP 2020*, pp. 9131–9143.

[20] S. Colton, A. Bundy, and T. Walsh. Automatic concept formation in pure mathematics. In *IJCAI 1999*, pp. 786–791.

[21] S. Colton and D. Wagner. Using formal concept analysis in mathematical discovery. In *Calculemus/MKM 2007*, LNCS 4573, Springer, 2007, pp. 205–220.

[22] K. Claessen, N. Smallbone, and J. Hughes. QuickSpec: guessing formal specifications using testing. In *TAP 2010*, LNCS 6143, Springer, 2010, pp. 6–21.

[23] M. Johansson, D. Rosén, N. Smallbone, and K. Claessen. Hipster: integrating theory exploration in a proof assistant. In *CICM 2014*, LNCS 8543, Springer, 2014, pp. 108–122.